Temporal Directed F-Hypergraphs for Hyperlink Prediction

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Abstract. We formulate temporal directed hyperlink prediction for citation systems represented as singleton-tail directed F-hypergraphs. A citing document is a tail and its contemporaneous bundle of cited documents is a directed hyperedge head. Given a partially observed head at time t, the task ranks a hidden head member against historically available candidates. The formulation separates event-level higher-order prediction from ordinary dyadic link prediction and makes temporal admissibility explicit. We define temporal directed F-hypergraph hyperlink prediction, establish the scope of unique-tail reconstruction, and specify a fixed-snapshot chronological evaluation design. The autonomous empirical study treats eight scoring methods on equal grounds: NHP-D-mean, NHP-D-maxmin+, Hyper-SAGNN, CHESHIRE, DHGNN, HPRA, co-citation, and historical in-degree. On identical frozen candidates in Supreme Court and European Court of Human Rights citation data, no method dominates every corpus and metric. NHP-D-maxmin+ is strongest on the SCOTUS ranking measures, while the smaller ECHR cohort displays metric-dependent results. The paper therefore offers a reproducible comparative benchmark for higher-order prediction on directed citation systems.

Keywords: directed hypergraphs, temporal networks, hyperlink prediction, graph neural networks, citation networks

1 Introduction

Many networked systems evolve through events that connect one source to a set of targets: a transaction invokes several instruments, a reaction maps reactants to products, and a document cites a bundle of earlier documents. A simple directed graph represents such an event by independent arcs. This is adequate for many graph-level questions, but it does not make the event head an explicit mathematical object on which a prediction task can be conditioned. Higher-order network models address this issue by representing interactions as hyperedges [1, 4]. The directed case is particularly important when source and target roles have distinct semantics.

Citation data provide a tractable real-world instance. Each opinion is a source node and its contemporaneous set of cited precedents is a directed head. The resulting object is a directed F-hypergraph, in the terminology of Gallo

et al. [4]. This paper focuses neither on legal interpretation nor on descriptive jurisprudence. It asks a network-science question: *can a temporally admissible directed-hypergraph model recover a hidden member of a citation-event head?*

The question is a higher-order analogue of link prediction. The prediction task now is conditioned on the remaining members of a set-valued head, candidates must predate the event, and graph-derived inputs cannot use later events. These requirements matter in any temporal directed-hypergraph application, not only in citation networks.

The contribution is fourfold. We (i) define temporal directed F-hypergraph hyperlink prediction and clarify the unique-tail case; (ii) specify a non-leaking chronological candidate-ranking design; (iii) give a common scoring interface and comparable descriptions for six published hypergraph-prediction methods plus two transparent baselines; and (iv) evaluate every method on the same frozen candidates, train–validation–test windows, and calibrated binary decision rule. The paper is deliberately narrow: it offers a mathematical task definition and an autonomous comparative benchmark rather than a theory of judicial institutions.

2 Temporal Directed F-Hypergraphs

Definition 1 (Temporal directed F-hypergraph). *Following [4], a directed hyperarc is an ordered pair (T, H) of disjoint tail and head subsets of $\mathcal{V}$. An* F-arc *has a singleton tail $T = \{u\}$ and is written (u, H). A temporal directed F-hypergraph is*

$$\mathcal{H} = (\mathcal{V}, \mathcal{E}, \tau), \qquad \mathcal{E} \subseteq \{(u, H) : u \in \mathcal{V},\ \emptyset \neq H \subseteq \mathcal{V} \setminus \{u\}\}, \tag{1}$$

where $\tau : \mathcal{V} \cup \mathcal{E} \to \mathbb{R}$ assigns times to vertices and F-arcs. An F-arc $e = (u, H)$ is time-respecting *when*

$$\tau(v) < \tau(e) = \tau(u), \qquad \forall v \in H. \tag{2}$$

For a citation event, u is the citing document and H is its set of cited precedents. The historical state immediately before time t is $\mathcal{H}_{<t} = (\mathcal{V}_{<t}, \mathcal{E}_{<t})$, where $\mathcal{V}_{<t} = \{v : \tau(v) < t\}$ and $\mathcal{E}_{<t} = \{e : \tau(e) < t\}$. Thus, condition (2) rules out a precedent that post-dates its citing opinion.

2.1 Dyadic projection and the unique-tail case

The directed dyadic projection is

$$\Pi(\mathcal{H}) = \{(u, v) : (u, H) \in \mathcal{E},\ v \in H\}. \tag{3}$$

The projection is informative, but its information-preservation properties depend on the event structure.

Proposition 1 (Unique-tail reconstruction). *Let $\mathfrak{F}_1$ be the class of directed F-hypergraphs in which each tail occurs in at most one F-arc. Then Π is injective on $\mathfrak{F}_1$.*

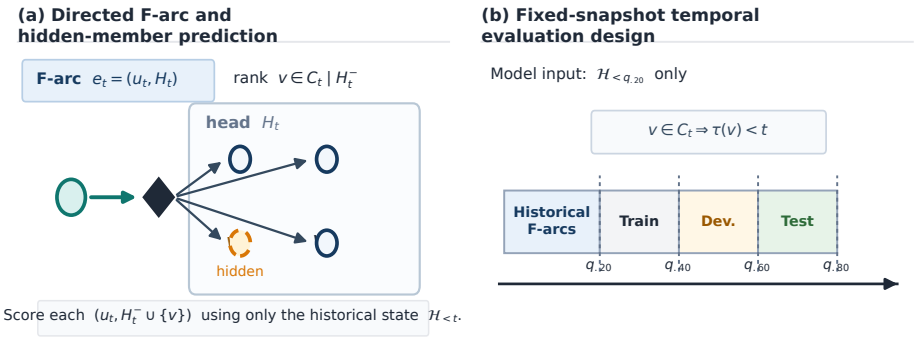


Fig. 1. Mathematical formulation and temporal evaluation design. (a) A directed F-arc has a singleton tail u_t and a set-valued head H_t; the prediction task hides $v^\star$. (b) The historical F-hypergraph snapshot precedes the train, validation, and test windows.

Proof. For each tail u, define $H_u = \{v : (u, v) \in \Pi(\mathcal{H})\}$. The unique-tail assumption implies that either $H_u = \emptyset$ or the sole F-arc with tail u is (u, H_u). Hence $\mathcal{E}$ is recovered from $\Pi(\mathcal{H})$. □

The present citation construction deliberately assigns each document one citation-list F-arc, so the premise of Proposition 1 holds. This resolves the apparent tension in the citation setting: the F-hypergraph is not claimed to contain extra raw information relative to an adjacency list with unique tails. Its role is to preserve an *event-level prediction task*: a candidate is scored together with the remaining cited head, not as an isolated dyadic arc. The non-injective case arises in applications in which the same source emits several events, such as recurring transactions, document revisions, or reactions; then the projection loses the partition into event heads.

3 Time-Respecting Hyperlink Prediction

For an observed F-arc $e_t = (u_t, H_t)$, choose $v^\star \in H_t$ and set $H_t^- = H_t \setminus \{v^\star\}$. Let $C_t \subseteq \mathcal{V}_{<t} \setminus H_t^-$ be a temporally admissible candidate set that contains $v^\star$. Temporal directed hyperlink prediction ranks

$$s\big(u_t, H_t^- \cup \{v\}; \mathcal{H}_{<t}\big), \qquad v \in C_t. \tag{4}$$

The task evaluates whether historical higher-order structure distinguishes the hidden member within a controlled candidate set; it does not ask a model to generate an unrestricted future citation list.

Figure 1 depicts the object and chronological evaluation design. A fixed historical snapshot is used: every graph input, random-walk embedding, and candidate popularity statistic is estimated before the train, validation, and test windows. Consequently, later F-arcs cannot alter a ranking at time t.

Algorithm 1 Fixed-snapshot temporal F-hyperlink ranking

Require: Time-respecting F-arcs $\mathcal{E}$; temporal cut points $q_{.20}, q_{.40}, q_{.60}, q_{.80}$
1: Build $\mathcal{H}_{<q_{.20}}$ and all historical candidate statistics
2: Initialize node features by Node2Vec on the snapshot clique expansion
3: **for** each event $e_t = (u_t, H_t)$ in the later windows **do**
4: deterministically hide $v^\star \in H_t$ and form $H_t^- = H_t \setminus \{v^\star\}$
5: form $C_t = \{v^\star\} \cup C_t^{\text{hard}} \cup C_t^{\text{rand}}$, with $|C_t| = 20$
6: score every candidate F-arc $(u_t, H_t^- \cup \{v\})$, $v \in C_t$
7: **end for**
8: fit on $[q_{.20}, q_{.40})$, select on $[q_{.40}, q_{.60})$, and report $[q_{.60}, q_{.80})$

3.1 Candidates, split, and ranking

Let $B_t = \{v \in \mathcal{V} : \tau(v) < t\}$ be the historically available vertex set. A valid candidate rule obeys

$$C_t \subseteq B_t \setminus H_t^-, \qquad v^\star \in C_t. \tag{5}$$

This excludes unavailable documents and prevents another positive head member from entering the negative pool. In every experiment, C_t has 20 candidates: the hidden true member, nine non-head nodes with highest historical in-degree, and ten uniformly sampled historical non-head nodes. The mixture creates both popularity-based hard alternatives and ordinary historical alternatives.

After chronological ordering, $q_{.20}$, $q_{.40}$, $q_{.60}$, and $q_{.80}$ denote the 20%, 40%, 60%, and 80% indices of the strictly time-valid F-arc sequence. The graph snapshot and all candidate statistics use $\mathcal{H}_{<q_{.20}}$ only. Events in $[q_{.20}, q_{.40})$ fit model parameters, those in $[q_{.40}, q_{.60})$ select the model, and those in $[q_{.60}, q_{.80})$ are held out. For each test event,

$$r_t = 1 + \sum_{v \in C_t \setminus \{v^\star\}} \mathbf{1}\big\{s(u_t, H_t^- \cup \{v\}; \mathcal{H}_{<t}) \geq s(u_t, H_t; \mathcal{H}_{<t})\big\}. \tag{6}$$

We report Recall@k (the proportion of hidden members ranked in the top k), MRR (the average reciprocal rank), mean rank (the average hidden-member position; lower is better), and candidate-row ROC-AUC. We also report F1, precision, recall, accuracy, log-loss, and Matthews correlation coefficient (MCC) after score calibration and threshold selection on the validation window only. Ties are resolved deterministically by candidate identifier for every method.

4 Models and Comparative Design

4.1 Candidate-scoring families

The study evaluates eight methods through one common prediction task $s_m(e_t, v)$, rather than treating any method as a reference. Two variants of the Neural Hypergraph Predictor for directed hypergraphs (NHP-D) are included: NHP-D-mean and NHP-D-maxmin+ [10]. For each candidate $v \in C_t$, NHP-D receives

$(u_t, H_t^- \cup \{v\})$. Node2Vec [5] initializes vertices from the fixed historical clique expansion, a graph convolution produces node states, and directed tail–head pooling produces hyperlink and direction logits.

Let $X \in \mathbb{R}^{|\mathcal{V}|\times d}$ be the 64-dimensional Node2Vec matrix and $\widetilde{A}$ the fixed historical clique-expansion adjacency with self-loops. The NHP-D node states are

$$Z = \sigma\Big(\widetilde{D}^{-1/2}\widetilde{A}\widetilde{D}^{-1/2}XW\Big), \tag{7}$$

where $\widetilde{D}$ is the degree matrix [6]. For a candidate F-arc (u, H), with $\bar{z}_H = |H|^{-1}\sum_{v\in H} z_v$, the singleton-tail local updates are

$$\ell_u = \rho(W_s z_u + W_n \bar{z}_H), \qquad \ell_v = \rho\left(W_s z_v + W_n \frac{z_u + \sum_{w\in H\setminus\{v\}} z_w}{|H|}\right). \tag{8}$$

NHP-D-mean uses $\psi_{\text{mean}} = (\ell_u + |H|^{-1}\sum_{v\in H}\ell_v)/2$, while NHP-D-maxmin+ uses $\psi_{\text{maxmin+}} = \rho(\ell_u - \min_{v\in H}\ell_v)$. Both optimize a hyperlink loss plus a weighted direction loss,

$$\mathcal{L} = \text{BCE}(y, \text{logit}_{\text{hyper}}) + 0.25\ \text{BCE}(y, \text{logit}_{\text{dir}}). \tag{9}$$

Table 1 describes each method at the same level: the historical structure it uses and the candidate-head score it supplies. The first six are published prediction families. HPRA [7], co-citation, and historical in-degree are not weaker versions of the neural models; they provide transparent structural comparisons that answer distinct questions about the value added by nonlinear or directed representation learning.

4.2 Common prediction interface and implementation audit

For every method m, the evaluated quantity is the same conditional candidate score,

$$s_m(e_t, v) = s_m\big(u_t, H_t^- \cup \{v\}; \mathcal{H}_{<q.20}\big). \tag{10}$$

Thus the methods differ in their structural map from the same fixed historical state to a candidate score, not in their temporal information, positives, negatives, or test events. The candidate F-arc is never inserted into the historical encoder. For HPRA, A is the historical degree-preserving head projection and D_v its vertex-degree matrix; its attachment score is

$$s_{\text{HPRA}}(v \mid H_t^-) = \sum_{w\in H_t^-}\big(A + AD_v^{-1}A\big)_{vw}. \tag{11}$$

All neural methods are trained on $[q.20, q.40)$ and selected by validation ranking on $[q.40, q.60)$. The NHP-D variants are trained anew under this same rule. Published model definitions and publicly available implementations were used to construct conditional adaptations to the common citation-ranking prediction; their differing native data formats are not allowed to change the frozen event or candidate-set specification. Each resulting score file was checked event by event: it contains one positive and 19 negatives, with candidate identifiers identical across methods in every split.

Table 1. Methods evaluated on the common evaluation design. Each receives the same frozen event, observed head, and candidate list.

Method	Historical representation	Candidate-head score
NHP-D-mean	Clique-expansion GCN with tail–head direction	Mean-pooled local tail/head states [10].
NHP-D-maxmin+	Clique-expansion GCN with tail–head direction	Max–min+ local tail/head pooling [10].
Hyper-SAGNN	Historical node embeddings	Diagonal-masked self-attention over $H_t^- \cup \{v\}$ [11].
CHESHIRE	Historical head-incidence features	Order-three Chebyshev convolution and max–min plus norm pooling [2].
DHGNN	Directed historical F-arc propagation	Directed embeddings and candidate–context dot product [9].
HPRA	Historical head projection	Resource-allocation attachment to H_t^- [7].
Co-citation	Historical head projection	Sum of direct head-association weights.
Historical in-degree	Historical head frequencies	$\log(1 + \text{head frequency}(v))$.

Table 2. Corpus and strict temporal-evaluation characteristics. The final cohort is smaller than the full graph because every head member and every candidate must be historically available under the fixed-snapshot evaluation design.

	SCOTUS	ECHR
Citation F-arcs in supplied graph	24,837	18,912
Strictly time-respecting F-arcs with $\lvert H\rvert \geq 2$	4,793	596
Historical snapshot F-arcs	958	119
Train / validation / test events	225 / 100 / 61	83 / 42 / 42
Held-out candidate rows	1,220	840
Candidates per event	20	20

5 Data and Experimental Design

The empirical instances are citation F-hypergraphs derived from Supreme Court of the United States (SCOTUS) opinions and European Court of Human Rights (ECHR) judgments. The SCOTUS corpus covers 1791–2024; ECHR is a 10,000-judgment English-language HUDOC sample covering 1959–2013. Citation networks have been used as longitudinal network data [3, 8]; here they are used only as directed higher-order event streams.

One head member is hidden deterministically per event. Event-level uncertainty is summarized by 2,000 bootstrap resamples. For the binary measures, each method's raw validation scores are standardized, calibrated by a one-variable logistic map, and assigned the threshold that maximizes validation F1; the fitted map and threshold are then applied once to the held-out scores. This leaves the rank order unchanged and makes F1, precision, recall, accuracy, log-loss, and

Table 3. Common evaluation contract. The evaluation design is shared by all methods, including the two projection controls.

Component	Specification
Historical representation	F-arcs strictly before $q_{.20}$; no later F-arc enters a graph-derived feature.
Temporal windows	$[q_{.20}, q_{.40})$ train; $[q_{.40}, q_{.60})$ validation; $[q_{.60}, q_{.80})$ test.
Candidate set	One hidden member, nine high-in-degree non-head candidates, and ten random historical non-head candidates.
Comparison unit	One directed candidate F-arc $(u_t, H_t^- \cup \{v\})$; 20 candidates per event.
Model selection	Validation-window ranking; held-out test window untouched until final scoring.
Uncertainty	2,000 event-level bootstrap resamples for Recall@1 and MRR.

MCC comparable despite different native score scales. The strict design makes each score a controlled temporal benchmark estimate, not a population estimate of all citations. The ECHR test cohort is particularly small; differences there must therefore be interpreted with its uncertainty intervals rather than as definitive method rankings.

5.1 Fitting configuration

The frozen evaluation design applies a single candidate construction and a single held-out cohort to all methods. NHP-D uses 64-dimensional Node2Vec features (20-step walks, ten walks per node, $p = q = 1$), AdamW optimization, and validation-window early stopping. Hyper-SAGNN, CHESHIRE, and DHGNN share the same historical vertices and candidate head. Each uses the common validation rule. Architecture-specific settings, score calibration, and binary thresholds are selected without inspecting test ranks or test labels. Changing either candidate construction or test-event eligibility would change the empirical question.

5.2 Calibration and metric interpretation

The native scores of the eight methods have different ranges and meanings: some are probabilities, others are logits, similarities, or resource-allocation values. A common binary threshold on those raw scales would not be meaningful. Therefore, for each method m and court separately, the validation scores are standardized and mapped to a probability by

$$p_m(y = 1 \mid s) = \left[1 + \exp\left\{-a_m \frac{s - \mu_m}{\sigma_m} - b_m\right\}\right]^{-1}, \tag{12}$$

where (μ_m, σ_m) and (a_m, b_m) are fitted from validation rows only. The binary threshold θ_m is the validation value that maximizes F1. Equation (12) and θ_m are then fixed before the test labels are used.

This calibration does not change a method's within-event rank ordering; Recall@k, MRR, mean rank, and ROC-AUC therefore retain their direct ranking interpretation. F1, precision, recall, accuracy, log-loss, and MCC instead assess the calibrated retain-or-reject decision. In the one-positive-to-19-negative candidate design, accuracy alone can be misleading. We consequently report it jointly with F1 and MCC. The F1 definition is

$$\mathrm{F1} = \frac{2\mathrm{TP}}{2\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}, \tag{13}$$

and MCC is the correlation of the binary predictions and labels, using all four confusion-matrix cells.

6 Results

The autonomous comparison yields no global winner. On SCOTUS, NHP-D-maxmin+ has the strongest Recall@1 (.410), Recall@5 (.770), MRR (.577), and candidate-row ROC-AUC (.824), whereas CHESHIRE has the strongest Recall@10 (.967). On ECHR, co-citation has the strongest Recall@1 (.214) and MRR (.344), Hyper-SAGNN has the strongest Recall@5 (.548), NHP-D-mean has the strongest Recall@10 (.857, tied with NHP-D-maxmin+ and Hyper-SAGNN), and NHP-D-maxmin+ has the highest ROC-AUC (.714). Table 4 therefore reports an empirical ordering by corpus and metric rather than assuming that the directed models must prevail.

Table 5 gives the requested binary evaluation. The validation-fitted thresholds test whether a candidate should be retained; this differs from its relative rank within an event. On SCOTUS, HPRA and co-citation have the highest F1 (.409) and MCC (.409), while NHP-D-maxmin+ has the smallest log-loss (.167). On ECHR, NHP-D-mean has the highest F1 (.161), while co-citation has the strongest precision (.308), accuracy (.944), and MCC (.148). These distinctions reinforce the need to report several measures.

Figure 2 displays MRR with 95% event-bootstrap intervals. It visualizes the principal ranking result without privileging a method a priori: SCOTUS favors NHP-D-maxmin+ on MRR, whereas the ECHR intervals are broad and overlap substantially. In conjunction with Tables 4–5, this is evidence for context-dependent comparative performance, not universal superiority.

Table 4. Autonomous chronological ranking evaluation on identical 20-candidate test lists. Boldface marks the best result within a court and column.

Court	Method	R@1	R@5	R@10	MRR	AUC
SCOTUS	Historical in-degree	.000	.000	.230	.078	.324
	Co-citation	.295	.377	.656	.385	.665
	HPRA	.295	.393	.639	.383	.665
	DHGNN	.328	.689	.738	.480	.767
	Hyper-SAGNN	.344	.738	.951	.530	.798
	CHESHIRE	.361	.754	**.967**	.522	.820
	NHP-D-mean	.311	.754	.934	.505	.785
	NHP-D-maxmin+	**.410**	**.770**	.951	**.577**	**.824**
ECHR	Historical in-degree	.000	.190	.333	.128	.378
	Co-citation	**.214**	.476	.667	**.344**	.689
	HPRA	.190	.476	.667	.324	.643
	DHGNN	.095	.476	.786	.273	.691
	Hyper-SAGNN	.119	**.548**	**.857**	.308	.699
	CHESHIRE	.095	.286	.643	.230	.591
	NHP-D-mean	.095	.524	**.857**	.302	.709
	NHP-D-maxmin+	.167	.476	**.857**	.323	**.714**

Table 5. Validation-calibrated binary test metrics on the same candidate rows. Thresholds maximize validation F1; lower log-loss is better.

Court	Method	F1	Prec.	Rec.	Acc.	AUC	Loss	MCC
SCOTUS	Historical in-degree	.166	.091	**1.000**	.499	.676	.186	.207
	Co-citation	**.409**	.594	.311	**.955**	.665	.177	**.409**
	HPRA	**.409**	.594	.311	**.955**	.665	.177	**.409**
	DHGNN	.263	**.667**	.164	.954	.767	.177	.316
	Hyper-SAGNN	.330	.405	.279	.943	.798	.172	.307
	CHESHIRE	.299	.500	.213	.950	.820	.174	.305
	NHP-D-mean	.291	.327	.262	.936	.785	.175	.260
	NHP-D-maxmin+	.265	.500	.180	.950	**.824**	**.167**	.280
ECHR	Historical in-degree	.127	.070	**.762**	.479	.622	.194	.099
	Co-citation	.145	**.308**	.095	**.944**	.689	.192	**.148**
	HPRA	.143	.286	.095	.943	.643	.193	.141
	DHGNN	.151	.086	.619	.651	.691	.188	.123
	Hyper-SAGNN	.132	.088	.262	.827	.699	.190	.073
	CHESHIRE	.062	.055	.071	.892	.591	.196	.006
	NHP-D-mean	**.161**	.094	.571	.702	.709	.185	.133
	NHP-D-maxmin+	.158	.099	.381	.796	**.714**	**.184**	.110

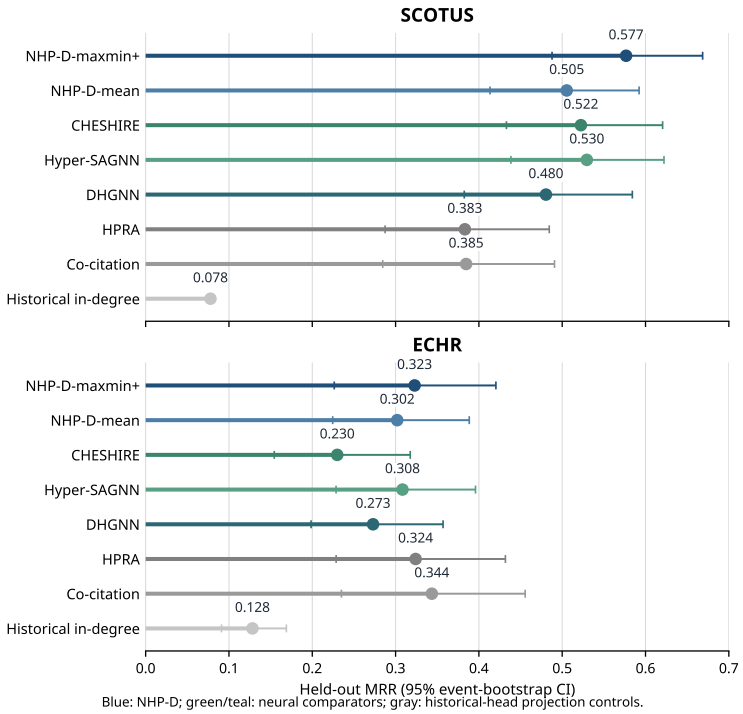


Fig. 2. Held-out MRR with 95% event-bootstrap intervals for all methods. Every score ranks the same hidden member against the same 19 negative candidates per event.

7 Validity Conditions and Limitations

The benchmark measures conditional ranking, not unrestricted generation of a future head. The partial head H_t^- is observed by construction and the model selects the hidden member from a finite historical candidate set. The design is valid only when three conditions hold: temporal validity requires (2); representation validity requires that graph-derived inputs use only the fixed historical snapshot; and event validity requires that candidate v be evaluated in the displayed F-arc $(u_t, H_t^- \cup \{v\})$ rather than as a projected pair in isolation.

The comparison has corresponding limits. Strict temporal eligibility leaves 61 SCOTUS and 42 ECHR test events, and the 20-candidate pool is sampled rather than exhaustive. Models with a head-only candidate architecture are evaluated on the displayed head; NHP-D and DHGNN additionally use directed historical roles. The implementations are documented conditional adaptations of published architectures to one citation-ranking prediction. A rolling snapshot could instead update representations after each event, but would introduce online retraining and candidate-refresh choices outside the present controlled comparison.

8 Discussion and Conclusion

Temporal directed F-hypergraph hyperlink prediction is well-defined only after the event boundary, historical candidate universe, hiding rule, and time-respecting feature construction are specified. The unique-tail proposition clarifies that a citation-list F-hypergraph need not add raw information beyond a complete adjacency list; its contribution is to preserve the set-valued event on which the prediction question is conditioned.

The comparative results establish a method-dependent empirical ordering rather than a preassigned winner. On SCOTUS, NHP-D-maxmin+ leads the principal ranking measures, while CHESHIRE leads Recall@10 and the projection methods lead some calibrated binary measures. On ECHR, co-citation leads Recall@1 and MRR, Hyper-SAGNN leads Recall@5, NHP-D-mean shares the best Recall@10 and has the best F1, and NHP-D-maxmin+ has the best ROC-AUC and log-loss. Thus, the contribution is an autonomous, reproducible temporal task and a comparative result showing how directed, set-aware, spectral, and projected representations behave on the same event-level question.

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